(Please write your Enrolment no. immediately) Enrl No_____

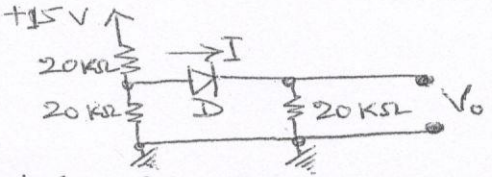
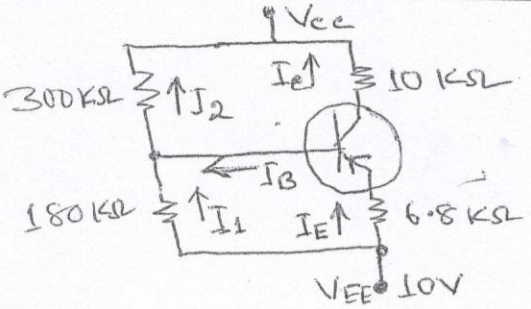
MID TERM EXAMINATION

B.Tech Programmes

Third Semester, November, 2022

Paper Code:-ECC-211 **Subject:-Analog Electronics-I**

Attempt Q No.1 which is compulsory and any two questions from remaining

Q.No	Question	M.M.	CO
1(a)	Determine the change in diode voltage Vd, corresponding to 15:1 change in diode current Id, for a junction diode operating at room temperature(Consider the diode to be Silicon diode)	2	CO1
1(b)	Distinguish between Diffusion capacitance and Transition Capacitance of Diode	2	CO1
1(c)	Derive and establish the relationship between α and β of transistor	2	CO2
1(d)	What is Early Effect in Common base configuration of BJT?	2	CO2
1(e)	Distinguish between Depletion type and Enhancement type MOSFET	2	CO2
2(a)	Differentiate between Avalanche and Zener breakdown	3	CO1
2(b)	 Draw thevenin equivalent of the above circuit and find out I and V_0	7	CO1
3(a)	Draw output characteristics of BJT in CE configuration, showing different regions of operations ,Load lines and Q-point	3	CO2
3(b)	 **Determine I_B, I_C, I_E, I_1, I_2, V_C, V_B and V_E**	7	CO2
4(a)	Deriving necessary equations, determine how the Stability factor **S(Ico)** is related to β for fixed-bias and self-bias circuits. Draw the graph of variation of **S(Ico)** with change in **RB/RE**	7	CO2
4(b)	Using Shockley's equation determine and draw the transfer curve of ID vs VGS if I_{DSS} is 18mA and Vp(pinch off voltage) is -6V.	3	CO2